Imaging of the Pediatric Breast: Review of Normal Development and Spectrum of Disease

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See the slide presentation here.

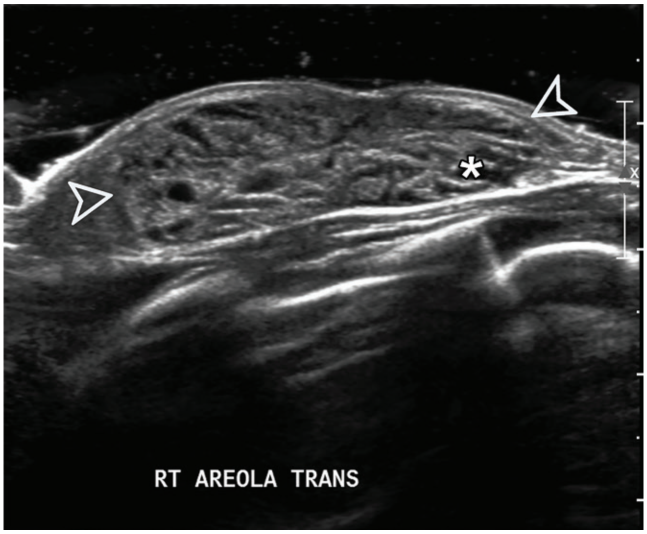

Figure 1. Enlarged right breast in a 10-day-old newborn girl. US image shows heterogeneous retroareolar tissue (arrowheads) in the right breast (greater than in the left breast, not shown), compatible with neonatal breast enlargement. Linear hypoechoic regions (*) represent branched ducts.

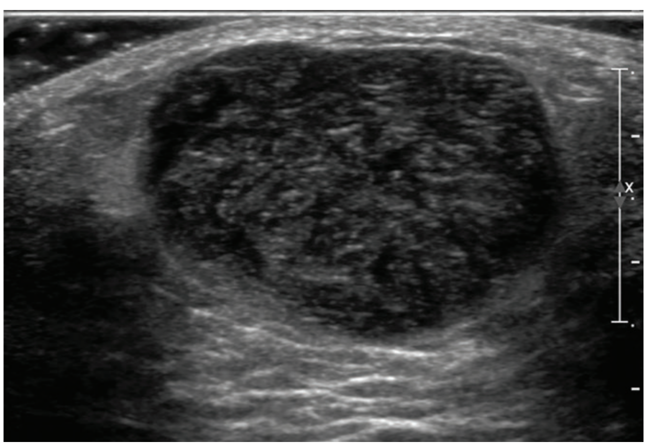

Figure 2. Fibroadenoma in a 13-year-old adolescent girl who presented with a palpable left breast mass. US image shows a mass that is well circumscribed, oval, parallel in orientation, and hypoechoic with posterior acoustic enhancement, findings compatible with a fibroadenoma.

Breast masses in the pediatric population are uncommon, with a prevalence of 3.2% in adolescent girls. However, when present they can be a source of significant stress for patients and families. The slide presentation reviews the imaging features of normal breast development and pathologic conditions.

A review of patient history and physical examination are essential for accurate initial assessment and diagnosis of pediatric breast masses. When necessary, imaging can play a helpful role in diagnosis, with US as the mainstay imaging modality. Mammography is not often used, given the risk of ionizing radiation to developing breasts and low sensitivity in the dense breast tissue of adolescents.

Familiarity with the clinical and sonographic appearances of normal breast development is essential for the radiologist performing pediatric breast evaluation to differentiate variants of normal development from pathologic conditions. The Tanner grading system is commonly used to describe both the clinical appearance of the developing breast and the appearance of breast tissue at US.

Neonatal breast enlargement is a benign proliferation of glandular tissue thought to occur in response to placental estrogens (Fig 1), typically occurring in the 1st week of life. In slightly older girls, premature thelarche is the early onset of breast development in prepubertal girls, typically before ages 7–8 years. Other developmental anomalies can present as a breast asymmetry, such as Poland syndrome, a rare abnormality characterized by congenital malformations of the chest wall.

The most common breast masses encountered in the pediatric population include simple cyst, retroareolar cyst, fibroadenoma, infection (including mastitis and abscess), hematoma, and fat necrosis. Fibroadenomas are extremely common, comprising 95% of excised breast masses in adolescents, and usually demonstrate typically benign imaging features such as circumscribed margins, oval shape, and parallel orientation (Fig 2).

Relatively uncommon but benign breast masses can also be seen in the pediatric population, including infantile hemangioma, Mondor disease, duct ectasia, juvenile fibroadenoma, pseudoangiomatous stromal hyperplasia (PASH), and intraductal papilloma. Malignant or potentially malignant breast masses in the adolescent population include phyllodes tumor and breast adenocarcinoma. Metastatic disease to the breast is much more common than primary breast adenocarcinoma in this population and has been

Slide Presentation

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TEACHING POINTS

- Discuss imaging modalities used to evaluate pediatric breast masses.
- Recognize the imaging features of physiologic neonatal breast development, normal breast development in adolescents, and variants of normal development.
- Review the spectrum of pediatric breast disease and describe appropriate management strategies.

reported in primary malignancies such as rhabdomyosarcoma, lymphoma, and leukemia.

The vast majority of adolescent breast masses are benign, meaning that US and/or clinical follow-up is generally appropriate. Tissue biopsy in adolescents should be carefully considered, given the risk of injury to the developing breast. Biopsy and surgical excision are reserved for highly suspicious lesions, such as those with suspicious imaging features or in patients with concerning findings on the history or physical examination. While there are widely used guidelines for breast imaging in adults, namely the Breast Imaging Reporting and Data System (BI-RADS), there are no evidence-based guidelines for management of pediatric breast masses. This leads to wide variability in management within and between institutions. As such, familiarity with normal development and the spectrum of common and uncommon benign and malignant masses in the pediatric population is essential to recognize pathologic conditions and guide management.

Pediatric breast masses, while uncommon, can be a source of distress for patients, families, and doctors, given the lack of evidence-based guidelines for diagnosis and management. For radiologists, knowledge of normal development and the imaging appearances of benign and malignant breast masses in the pediatric population is paramount for accurate recognition of disease and appropriate management.

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Suggested Readings

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